

- 1 Fig. 1.1 is a transmission electron micrograph of a cell from the root of thale cress, *Arabidopsis thaliana*.

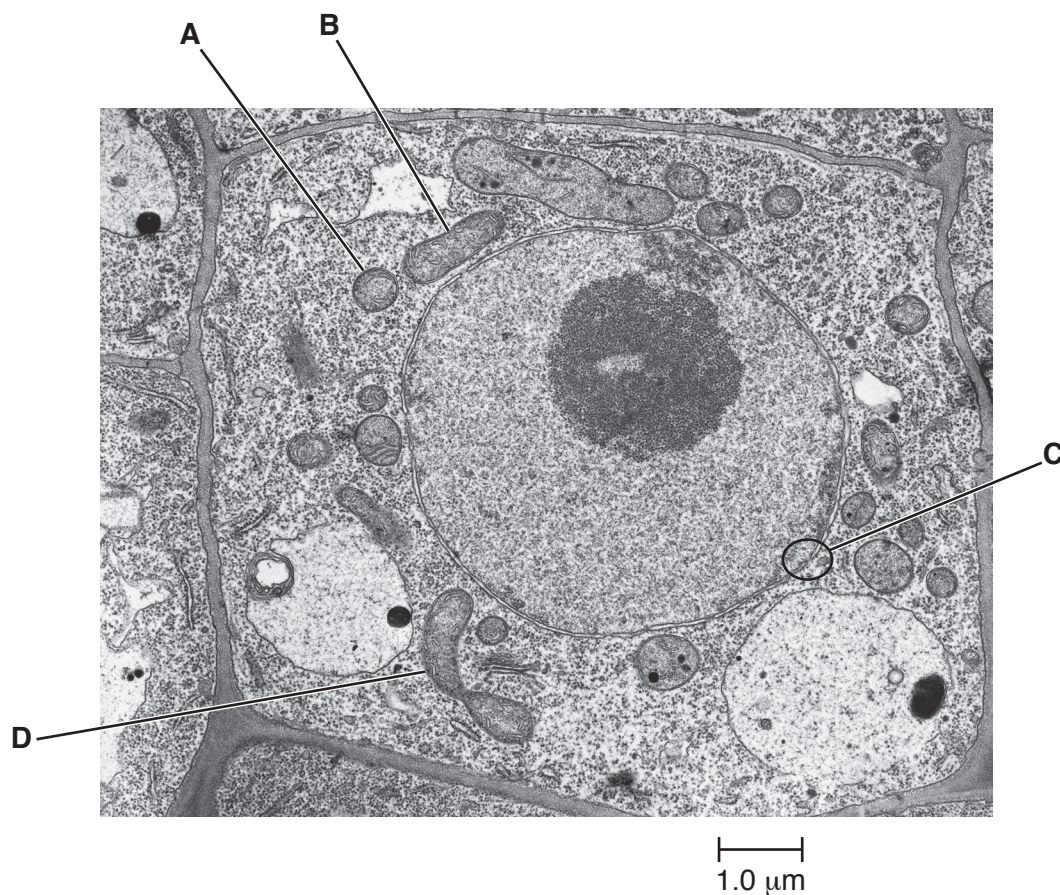


Fig. 1.1

- (a) (i) The structures labelled **A** and **B** on Fig. 1.1 are sections of two mitochondria. Suggest why **A** and **B** are different shapes.

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.....[1]

- (ii) The structure labelled **D** on Fig. 1.1 is a mitochondrion about to divide. Explain the importance of the division of mitochondria for the cell shown in Fig. 1.1 and for cells in the root tips of thale cress.

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.....[2]

- (b) Within a cell, substances move between the nucleus and the cytoplasm. The area labelled **C** in Fig. 1.1 shows an area where this communication occurs.

Make a large, labelled drawing of area **C** to show where this communication occurs.

[2]

- (c) Outline the functions of the nucleus in non-dividing cells, such as the cell in Fig. 1.1.

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[Total: 9]